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IMPORTANT INFORMATION

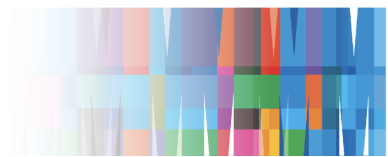
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Introduction

Lecture 1

ETF2480 & ETF5248 - S2, 2026
Jessica Leung, PhD, CAP-X, SFHEA

Today's Goal



- Admin
- What is Optimisation?
- Linear Programming: what and how?
- Some Linear Algebra
- Useful Transformation

Seminars, Workshops, Tutorials and Communications

Seminars & Workshops: Tuesdays, 12nn-3pm

Tutorials: Tutorials are on Mondays and Thursdays. Please refer to your timetable for the exact time and location.

Moodle Site: Let's have a look on Moodle.

We are using **Ed Discussion Board** for announcements, discussion, coding syntax and tutorial guide!

Teaching Team

Chief Examiner/Lecturer

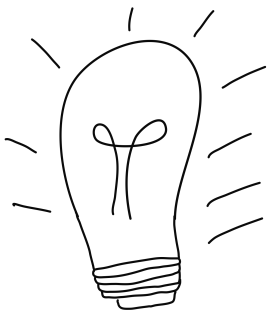
Dr Jessica Leung
Consultation: Tuesday 10:00am
Email: jessica.leung@monash.edu

Tutors

Ms Ziyi Deng - **Email:** ziyi.deng1@monash.edu
Ms Tina Jafari - **Email:** tina.rashidjafari@monash.edu
Mr Hamed Vosgha - **Email:** hamed.vosgha@monash.edu

You can also find our contact details on Moodle.

Textbook and Software



You don't need to buy textbook$.

There are no required textbooks in this unit. However, the following texts can serve as auxiliary or reference texts.

- D. Bertsimas and J. N. Tsitsiklis, **Introduction to Linear Optimization**.
- W. L. Winston, **Operations Research: Applications and Algorithms**.
- B. Taylor, **Introduction to Management Science: Global Edition**.
- R. J. Vanderbei, **Linear Programming: Foundations and Extensions**.

We teach in R and support Python.

We'll be using R as a tool to solve optimisation problems in this course.

- open source (like all real languages)
- can be used to write simple scripts but also is very efficient
- supports really fancy stuff (that we won't use in this course)
- two of the most popular languages for data analytics and data science
- We will start from the basics, with templates to follow
- We are also very happy to **support intermediate Python users using the Gurobi interface and packages such as PuLP.**

More information in the first tutorial and on Moodle.

The Roadmap: What is this unit about?

Topics to cover

- LP Basics and Examples (Week 1-4)
- Duality and Sensitivity (Week 5-7)
- Network Flow Problems (Week 7-8)
- (Mixed) Integer Programming (Week 9-10)
- Advanced Modelling (Week 11)
- Review (Week 12)

Assessments

Assessment task	Weight	Due date	Group or individual	CC or AI-enable?
Assignment 1	15%	Week 4	Individual	Both
Mid Semester Test	20%	Week 6 Seminar Time	Individual	CC
Assignment 2	15%	Week 10	Individual	Both
Final Exam	50%	Final Exam Period	Individual	CC

Table: Assessment Summary

Note: "CC" stands for Core Competency. This designation signifies that students are expected to demonstrate mastery of these tasks independently, without the use of any Generative AI tools.

AI transforms entry level work

Here is a quote from 2026 AI Global Jobs Barometer by PwC: “The traditional career ladder is compressing. AI-exposed junior roles are seven times more likely to demand traditionally senior skills. While overall early-career job postings have flatlined in highly AI exposed sectors, ’seniorised’ entry level roles are thriving, showing 35% growth since 2019."

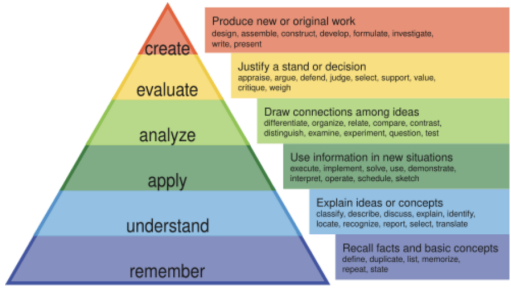


Figure: Bloom’s Taxonomy (Revised). The picture is by Tidema - Own work, CC BY 4.0, https://commons.wikimedia.org/w/index.php?curid=152872571. This is a hierarchical model for cognitive skills. You cannot analyze a concept if you do not understand it, and you cannot create something new without evaluating existing data.

Any Questions?

Do you have any questions about the admin side of things?

What is Optimisation in Business?

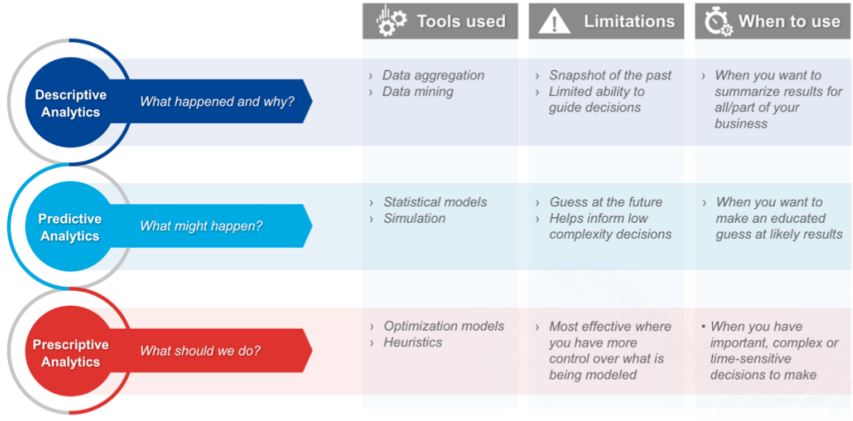
So, what is Optimisation in Business?

Hmmm...perhaps we should start with...

What is Business Analytics?

"**Business analytics** can be viewed as the intersection of a variety of disciplines, of which Operations Research, machine learning, and information systems are of particular relevance . As a process, it can be characterised by **descriptive, predictive**, and **prescriptive** model building using heterogeneous and 'big' data sources."

Hindle et al 2020, European Journal of Operations Research



Optimisation in Business

Represent your complex business problems as mathematical models, which you can adjust to accurately reflect your company's present-day reality. Use those models to help you tackle your real-world business problems and make the best possible decisions.

1. What business problem are you facing? What do you get to decide?
2. Formulate a mathematical optimisation problem that models your decision problem.
3. Solve the optimisation problem and make the decision.

Interesting read:

1. **How a Mathematical Optimization model can help your business deal with disruption** (Forbes 2020): Link to article
2. **Business analytics: Defining the field and identifying a research agenda** (EJOR 2020): Link to article

Linear Programming: what and how?

What is linear programming?

It is a type of mathematical optimization **problem**.

So there is no programming?

We'll use some R or Python code to solve LPs.

A mathematical optimization problem

$$\begin{aligned} \text{Maximize } & f(x) \\ \text{subject to } & g(x) \leq b \\ & x \geq 0 \end{aligned}$$

Objective Function:
A function we want to optimize (maximise or minimise)

Decision Variables:
Variables whose values are under our control

Constraints:
Restrictions on the decision variables values

$\rightarrow$ When $f(x)$ and $g(x)$ are both linear, we have an LP.

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Restrictions on the decision variables values
$\rightarrow$ When $f(x)$ and $g(x)$ are both linear, we have an LP.

Linear Programming (LP)

with 2 variables

$$\begin{aligned} \text{Maximize } & c_1 x_1 + c_2 x_2 \\ \text{subject to } & A_{11} x_1 + A_{12} x_2 \leq b_1 \\ & A_{21} x_1 + A_{22} x_2 \leq b_2 \\ & x_1, x_2 \geq 0 \end{aligned}$$

Simple Example

ApplePie is a computer retail store that sells desktops and laptops.

The company earns $600 on each desktop computer it sells and $900 on each laptop.

Due to COVID19 supply chain disruption, the supplier of ApplePie cannot ship more than 80 desktops and 75 laptops to ApplePie next month.

The employees at ApplePie must spend 2 hours installing software and checking each desktop computer they sell. They spend 3 hours to complete this process for laptop computers. They expect to have about 300 hours available for this purpose during the next month.

The store's management is fairly certain that they can sell all the computers they order, but are unsure how many desktops and laptops they should order to **maximize profits**.

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LP formulation

Let x_1 and x_2 be the number of desktop computers and laptop computers ApplePie order respectively.

$$
\begin{aligned}
\text{Maximize } & 600x_1 + 900x_2 \\
\text{subject to } & 2x_1 + 3x_2 \leq 300 \\
& x_1 \leq 80 \\
& x_2 \leq 75 \\
& x_1, x_2 \geq 0
\end{aligned}
$$

(Some) Linear Algebra

What do you know about the matrix?

The movie?

No, the rectangular array of numbers that can be used to express an LP.

We need some tools to solve the LPs.

A (gentle) introduction to vector, matrix, some linear algebraic operations and how do we use them to represent an LP. We will see how it all makes sense in R (and Python).



Vector

A vector of size n (or n-vector)

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

We can also write

$$x = (x_1, x_2, \ldots, x_n).$$

x_i is the i-th **element/ component/ entry** of x.

These entries are real numbers unless specified otherwise.

Sometimes you may see us write

$$x \in \mathrm{R}^n$$

Matrix

A matrix of size $m \times n$ (m by n)

$$A = \begin{bmatrix} A_{11} & A_{12} & \cdots & A_{1n} \\ A_{21} & A_{22} & \cdots & A_{2n} \\ \vdots & \vdots & & \vdots \\ A_{m1} & A_{m2} & \cdots & A_{mn} \end{bmatrix}$$

A_{ij} is the i-th row j-th column **component/ element/ entry** of A.

You can view matrix as a bunch of vectors stick together.

Sometimes you may see us write

$$A \in \mathrm{R}^{m \times n}$$

Operations

Inner product of n-vectors

$$c^T x = c_1 x_1 + c_2 x_2 + \ldots + c_n x_n$$

is the **sumproduct** of every entry in both vectors. That means we can express our linear **objective function** in a compact way.

Matrix-vector multiplication

$$Ax = \begin{bmatrix} A_{11} & A_{12} & \cdots & A_{1n} \\ A_{21} & A_{22} & \cdots & A_{2n} \\ \vdots & \vdots & & \vdots \\ A_{m1} & A_{m2} & \cdots & A_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} A_{11}x_1 + A_{12}x_2 + \cdots + A_{1n}x_n \\ A_{21}x_1 + A_{22}x_2 + \cdots + A_{2n}x_n \\ \vdots \\ A_{m1}x_1 + A_{m2}x_2 + \cdots + A_{mn}x_n \end{bmatrix}$$

That means we can express our linear **constraints** in a compact way.

$$\begin{aligned} A_{11}x_1 + A_{12}x_2 + \cdots + A_{1n}x_n &\leq b_1 \\ A_{21}x_1 + A_{22}x_2 + \cdots + A_{2n}x_n &\leq b_2 \\ &\vdots \\ A_{m1}x_1 + A_{m2}x_2 + \cdots + A_{mn}x_n &\leq b_m \end{aligned}$$

$$\downarrow$$

$$Ax \leq b$$

So, the ApplePie LP can be represented like this...

LP formulation

Let x_1 and x_2 be the number of desktop computers and laptop computers ApplePie order respectively.

$$\begin{aligned} \text{Maximize } & 600x_1 + 900x_2 \\ \text{subject to } & 2x_1 + 3x_2 \leq 300 \\ & x_1 \leq 80 \\ & x_2 \leq 75 \\ & x_1, x_2 \geq 0 \end{aligned}$$

The matrix-vector form helps us code in R!

LP formulation

Maximize $c^T x$

subject to $Ax \leq b$

$$x \geq 0$$

$$c = \begin{bmatrix} 600 \\ 900 \end{bmatrix} \quad b = \begin{bmatrix} 300 \\ 80 \\ 75 \end{bmatrix}$$

$$A = \begin{bmatrix} 2 & 3 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

R code to solve the Applepie LP

```
library(lpSolve)

c <- c(600, 900)

A <- matrix(c(2,3,1,0,0,1), nrow=3, byrow=T)

b <- c(300,80,75)

res <- lp("max", c, A, "<=", b)

res

res$solution
```

The same goes with Python!

Please refer to this page for explanation of the code

Python code to solve the Applepie LP

```
import gurobipy as grb
import numpy as np

c = np.array([600, 900])
A = np.array([[2,3],[1,0],[0,1]])
b = np.array([300,80,75])

M = grb.Model('Applepie')
x = M.addMVar(2)
M.setObjective(c@x, grb.GRB.MAXIMIZE)
M.addConstr(A@x <= b)
M.optimize()
print(M.objVal)
print(x.x)
```

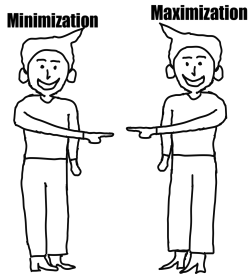
Compact and scalable ApplePie LP

Now imagine Applepie offers 3000 products.

- How would the dimensions of the matrix and vector change?
- Does the matrix-vector mathematical representation of the LP change?
- How would the coding syntax change?

Useful Transformations

There are more than one way to express an LP.



A max problem can be solved as a min problem.

We can solve the **maximisation problem**

$$\begin{aligned} \text{maximize} \quad & c^T x \\ \text{subject to} \quad & Ax \leq b \\ & Dx = f \end{aligned}$$

by minimising the function $-c^T x$ subject to the same constraints:

$$\begin{aligned} \text{minimize} \quad & -c^T x \\ \text{subject to} \quad & Ax \leq b \\ & Dx = f \end{aligned}$$

Equivalent problems

Two problems are (informally) **equivalent** if the solution of one is readily obtained from the solution of the other, and vice-versa.
Transforming maximisation problem into minimisation is a good example (see previous slide).
Some common transformations:

- **converting "$\leq$" constraints to "$\geq$" constraints:**

$$a_i^T x \leq b_i \quad \text{is equivalent to} \quad (-a_i)^T x \geq -b_i$$

- **eliminating equality constraints:** $a_i^T x = b_i$ is equivalent to two inequality constraints

$$(-a_i)^T x \geq -b_i \quad \text{and} \quad a_i^T x \geq b_i$$

LP in inequality form

Because all constraints are linear, only three cases are possible:

- $a_i^T x \leq b_i$
- $a_i^T x \geq b_i$
- $a_i^T x = b_i$

We can convert all constraints to $\geq$ form:

- $a_i^T x \leq b_i$ can be rewritten as $(-a_i)^T x \geq -b_i$
- $a_i^T x = b_i$ is equivalent to two inequality constraints

$$(-a_i)^T x \geq -b_i \quad \text{and} \quad a_i^T x \geq b_i$$

Hence, any LP can be written as

$$\begin{aligned} \text{minimize} \quad & c^T x \\ \text{subject to} \quad & Ax \geq b \end{aligned}$$

Example

$$\begin{aligned} \text{minimize} \quad & 2x_1 - x_2 + 4x_3 \\ \text{subject to} \quad & x_1 + x_2 + x_4 \leq 2 \\ & 3x_2 - x_3 = 5 \\ & x_3 + x_4 \geq 3 \\ & x_1 \geq 0 \\ & x_3 \leq 0 \end{aligned}$$

in matrix notation

$$\begin{aligned} \text{minimize} \quad & c^T x \\ \text{subject to} \quad & Ax \geq b \end{aligned}$$

where

$$A = \begin{bmatrix} -1 & -1 & 0 & -1 \\ 0 & 3 & -1 & 0 \\ 0 & -3 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix}, \quad b = \begin{bmatrix} -2 \\ 5 \\ -5 \\ 3 \\ 0 \\ 0 \end{bmatrix}, \quad c = \begin{bmatrix} 2 \\ -1 \\ 4 \\ 0 \end{bmatrix}.$$

Quick Summary

1. Identify the decisions you have to make (decision variables)
2. Formulate an optimisation problem for the business scenario (objective function and constraints).
3. Solve the optimisation problem (with R and report the result).

Linear Programming (LP) formulation

$$
\begin{aligned}
&\text{Maximize } c^T x \\
&\text{subject to } Ax \leq b \\
&\qquad\qquad x \geq 0 \\
&\qquad\quad \Updownarrow \\
&\text{Minimize } -c^T x \\
&\text{subject to } Ax \leq b \\
&\qquad\qquad x \geq 0
\end{aligned}
$$

b is a m-vector
c is a n-vector
x is a n-vector
A is a $m \times n$ matrix.

R code to solve the LP

```
library(lpSolve)
c <- c(...)
A <- matrix(c(...), nrow=..., byrow=T)
b <- c(...)
res <- lp("max", c, A, "<=", b)
res
res$solution
```

Workshop: Check your understanding (Attempt: 10 mins)

Steve's online shop

Steve has an online shop where he sells hand made candles and diffusers that uses natural and high quality essential oil. He sells the candle for $20 and the diffusers for $40. It takes him, on average, 20 minutes to prepare 1 candle and 35 minutes to prepare 1 diffuser. Steve has a day job and he only run the online business when he has time. He cannot spend more than 8 hours a week to make candles and diffusers. Additionally, he should make not more than 50 candles and 10 diffusers per week. It costs him $15 dollars to make each candle and $18 dollars to make each diffuser. How many candles and diffusers should he make each week to maximize his profit. Formulate and solve the LP to Steve model his business.

Steve's online shop: LP formulation

Live Coding Time! (Attempt: 10 mins)

Let’s code Steve’s online shop problem together!

Take home: Check your understanding

Giapetto's Woodcarving, Inc., manufactures wooden soldiers and trains.
Each soldier built:

- Sells for $27 and uses $10 worth of raw materials.
- Increases Giapetto's variable labor/overhead costs by $14.
- Requires 2 hours of finishing labor.
- Requires 1 hour of carpentry labor.

Each train built:

- Sells for $21 and uses $9 worth of raw materials.
- Increases Giapetto's variable labor/overhead costs by $10.
- Requires 1 hour of finishing labor.
- Requires 1 hour of carpentry labor.

Each week Giapetto can obtain:

- All needed raw material.
- Only 100 finishing hours.
- Only 80 carpentry hours.

Demand for the trains is unlimited. At most 40 soldiers are bought each week. Giapetto wants to maximize weekly profit.

Formulate an LP and solve using R.